

APPLIED RESEARCH

Machine Learning for Early Detection of Phishing URLs in Parked Domains: An Approach Applied to a Financial Institution

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ABSTRACT Phishing attacks remain a critical threat in the digital era, exploiting social engineering tactics to compromise user trust and sensitive information, often resulting in financial loss and identity theft. These attacks typically exploit multiple communication channels to impersonate trusted entities, inducing users to click on and access malicious Resource Locations (URLs). Early detection of these URLs can considerably reduce the volume and impact of these attacks. This paper proposes a machine learning-based framework for early detection of phishing URLs, specifically addressing newly registered and parked domains. Using a curated dataset comprising 211,659 URLs obtained from real-time SSL (Secure Sockets Layer) certificate monitoring, popular domain listings, and phishing incident reports, the methods encompass data pre-processing, feature engineering, and model optimization. A Light Gradient Boosting Machine classifier achieved recall of 96.02% and accuracy of 97.28% on a balanced dataset, validated through 10-fold cross-validation. Additionally, feature selection techniques reduced model complexity while maintaining detection effectiveness, facilitating practical deployment. The proposed frameworks provide a versatile tool for phishing prevention and brand protection, demonstrating potential applicability in sectors such as banking and e-commerce. This work contributes to proactive cybersecurity practices addressing evolving threats within digital ecosystems.

INDEX TERMS Phishing, machine learning, anti-fraud, brand protection, feature selection.

I. INTRODUCTION

The advent of the digital age, characterized by the pervasive integration of the internet into everyday life, has fundamentally altered societal interactions. While e-commerce, digital communication and payment systems offer significant conveniences, they concurrently introduce novel security challenges. The ease with which individuals can conduct online transactions and share information creates an attractive ground for cybercriminal activities. Social engineering

attacks, such as phishing, exploit these conveniences to deceive users and illegally obtain personal and financial data.

A phishing attack can be defined as a fraudulent digital communication where cybercriminals impersonate a trusted individual or organization to deceive, coerce, or induce the target into performing a specific action, typically providing sensitive information or downloading malicious software [1].

Studies conducted by the Anti-Phishing Working Group (APWG), based on the analysis of reports received over a given period, demonstrate that the financial and payment industries worldwide were targets of 31.5% of email phishing attempts in the fourth quarter of 2023, a figure very close to that observed in 2022 (33.7%). The analysis of recent reports

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also reveals a growing trend of voice phishing, using phone calls, SMS, or email, reflecting the emergence of hybrid phishing attacks that involve approaching victims through multiple channels. In the third quarter of 2024, APWG observed 932,923 phishing attacks, up from 877,536 in the second quarter, highlighting a continuous rise in phishing incidents. Social media platforms were once again the most frequently attacked sector, representing 30.5% of all phishing attacks [2]. These attacks typically involve a URL sent to the user, aiming to prompt them to enter sensitive information or download a malicious file.

In the context of digital payments industry, cybercriminals impersonate financial institutions or other trusted entities to trick users into disclosing confidential or critical information, strategically aiming to steal online identities and acquire financial benefits. The identities of a user online include usernames, passwords (for example, for online banking), credit/debit card numbers, PINs, card verification codes (CVVs) and expiration dates; compromising this data, phishers can perpetrate online financial theft by impersonating the victim [3]. Phishers steal user identities through various contact methods, such as impersonating a financial institution employee and using phone calls, emails, SMS, also in social media, such as messaging apps to deliver malicious links that trick users into providing critical data or downloading malicious software [4].

The adaptability of phishing across multiple platforms also remains as a relevant threat vector for new technologies in cybersecurity contexts, such as Vehicle-to-Everything (V2X) communication. Recent research highlights that phishing mechanisms have the potential to target vehicular communication networks, exploiting the inherent trust within these systems to deceive users and compromise security [5].

In the current cyber threat landscape, traditional phishing detection methods, such as deny-listed-based approaches that rely on user reports and expert evaluations, are becoming increasingly ineffective as cyber adversaries continuously refine their tactics. The rapid evolution of cybercriminal techniques, including the algorithmic generation of new domains and the ease of domain manipulation, requires the adoption of more proactive and adaptive detection strategies. Using advanced techniques, such as behavioral analysis and machine learning, it is possible to develop more robust defense mechanisms to mitigate and prevent cyberattacks [6], [7]. Such a comprehensive approach mitigates the inherent limitations of human vigilance, providing a proactive defense against phishing attacks by combining automated systems for early detection of fraudulent websites by user alerting and proactive countermeasures, can be an effective complement for user education initiatives.

Several studies have focused on applying machine learning techniques to detect phishing, often targeting fraudulent emails, SMS, or active websites [3], [7], [8], [9]. However, in the broader context of brand protection and customer-side fraud, a significant gap remains in the development of targeted solutions for the banking sector and

e-commerce, particularly concerning early detection. Most existing approaches identify suspicious URLs after a phishing attempt, when it has been reported or an incident has already occurred, meaning the malicious site is often already active and potentially impacting users. Our research proposes a process designed to proactively identify and flag URLs that mimic specific financial institutions as soon as they are registered, bringing them forward for analysis. The primary contribution of this work is a machine learning-based framework for the early detection of phishing URLs, specifically targeting newly registered and parked domains. A secondary contribution is a comparative performance evaluation of this framework across various machine learning algorithms.

The methods have been validated in a real-world scenario, where we continuously monitored domains targeting a prominent Brazilian bank, demonstrating its effectiveness in detecting phishing threats.

The structure of this paper is organized as: Section II provides a review of related work, offering insights into existing approaches and identifying ongoing challenges in the field of phishing detection related to the banking industry. Section III describes the materials and methods used to collect and prepare the database and develop a binary classifier, focusing particularly on recall and accuracy metrics to select the most effective models, namely Extreme Gradient Boosting (XG-Boost), Light Gradient Boosting Machine (LGBM), Logistic Regression and Random Forest Classifier (RFC). In Section IV, we present and analyze the results, highlighting the performance of each model across different stages. Finally, Section V concludes the paper by summarizing the key findings.

II. RELATED WORKS

This section investigates the application of machine learning algorithms in the cybersecurity domain, focusing specifically on detecting phishing URLs. The initial analysis involved reviewing relevant articles retrieved from Scopus, IEEE Xplore, and the ACM Digital Library, published between 2022 and 2024, on phishing detection using machine learning techniques and their contributions. Subsequently, the review examined key characteristics studies specifically applied to URL-based phishing detection using Machine learning, including the datasets utilized, feature selection and dimensionality reduction techniques. The presentation is subdivided into 2 subsection, the first one presents a comprehensive review of recent research advancements in phishing detection using machine learning techniques. The second section focus on specific recent methodologies that leverage URL-based phishing detection dataset and features.

A. MACHINE LEARNING FOR PHISHING DETECTION

This section provides a comprehensive background and review of recent research advancements in phishing detection using machine learning techniques. Phishing, a pervasive threat in the digital landscape, constitutes a category of cyberattacks that exploit human vulnerabilities by

manipulating decision-making processes [7]. These attacks target a wide range of organizations, including financial institutions, e-commerce platforms, and government agencies, along with their users or customers [6]. Fundamentally, phishing is a social engineering attack where malicious actors impersonate trusted entities to deceive victims into divulging sensitive information. This often involves the creation of fraudulent websites that mimic legitimate ones (e.g., login pages) to trick users into entering credentials or downloading malicious software that can capture and disclose data [10]. It may also attempt to leverage fraudulent emails, instant messages, and phishing techniques to redirect users to malicious websites [11].

A recent review on phishing taxonomy for deep learning-based detection methods identified six categories of phishing based on the media used, as we related in the Table 1. Website phishing involves fake web pages that mimic legitimate brands or individuals by abusing their images or credentials. While email phishing includes various subtypes, such as spear phishing, which targets specific individuals or groups, and whaling, which focuses on high-profile executives. SMS phishing uses malicious text messages to trick victims into clicking harmful links or sharing sensitive information. Voice phishing relies on phone calls, where attackers impersonate trusted entities to deceive victims. Mobile device phishing exploits vulnerabilities in smartphones, taking advantage of users' growing dependence on these devices. Finally, online social network phishing leverages platforms similar to Facebook and Instagram to impersonate trusted contacts. The study emphasizes that each type of phishing presents unique challenges and requires specific detection techniques for effective identification [4].

TABLE 1. Phishing taxonomy based on media type (Do et al. [4]).

Phishing Category	Description/Examples
Website	Fake webpages with brand images (e.g., bank login pages).
Email	Deceptive emails, including spear phishing and whaling.
SMS (Smishing)	Malicious links in SMS (e.g., fake delivery notices).
Voice (Vishing)	Impersonation via phone calls (e.g., fake bank calls).
Mobile Device Phishing	Malicious apps or device vulnerabilities (e.g., fake app stores).
Social Network Phishing	Impersonating contacts on platforms (e.g., Facebook, Instagram).

Researchers have proposed a variety of anti-phishing techniques to combat the growing threat of phishing attack, that can be classified into five main categories: deny listed-based systems that rely on known phishing sites, rule-based systems that use predefined rules to identify malicious content, third-party reporting systems that rely on user-submitted reports, visually similarity systems that compare websites to known phishing examples and machine

learning-based systems that use algorithms to learn and identify phishing patterns [12].

Machine learning algorithms have gained significant attention in cybersecurity due to their ability to analyze vast amounts of data in real-time, detecting anomalous patterns in network traffic, user behavior, and system logs. By proactively mitigating risks, these algorithms can identify novel and complex cyber threats that traditional rule-based systems may overlook, enhancing the efficacy of cybersecurity protocols [13]. Our analysis reveals developments of machine across five main approaches of phishing detection: Phishing Susceptibility Models, Website/URL Phishing Detection, Email Phishing Detection, SMS Phishing Detection (Smishing), and Voice Phishing Detection (Vishing), as summarized in Table 2.

As categorized in Table 2, some studies have analyzed phishing from a human-centric perspective, proposing Phishing Susceptibility Models (PSM), identifying several demographic and psychological factors that influence an individual's susceptibility to phishing attacks. Their finds suggest younger users, particularly those aged 18-25, are more likely to fall victim than those aged between 25-50 years old. Gender is also a contributing factor, with women mostly exhibiting higher susceptibility rates than men. Individuals with computer science backgrounds may possess a slight advantage in recognizing and avoiding phishing attempts compared to those with other educational backgrounds. Additionally, anti-phishing education, particularly when delivered after a firsthand experience, can significantly improve users' awareness and resistance to phishing attacks [7], [14].

Phishing susceptibility is categorized into long-term stable factors (e.g., knowledge, personality), situational factors (e.g., stress, workload), and in-the-moment factors (e.g., cognitive effort, email presentation). The model describes interactions among these factors, addressing the critical role of context in phishing vulnerability [7].

Yang et al. [15] proposed a Multidimensional PSM. Their research constructed a dataset with two types of e-mails. First, a set of 7 e-mails comprising 3 legitimate and 4 phishing emails were sent to 1,105 volunteers aged between 20 and 50 years. Demographic, personality, knowledge, experience, security behavior, and cognitive process data were gathered from each participant through a questionnaire. Seven supervised learning methods were employed to classify these volunteers into two categories: susceptible to phishing and not susceptible to phishing. Findings from such models can contribute to preventing phishing, suggesting more suitable information security awareness and training program [15].

Several studies have explored the application of machine learning techniques to detect phishing attempts or incidents. Some researches, for instance, developed e-mail phishing detection, as shown in Table 2. In this approach, the aim is weather spam detection or phishing attack prediction using natural language processing (NLP) techniques [16]. A recent study, for instance, leverages Long Short-Term Memory

(LSTM) neural networks to process and classify phishing emails in large datasets. Their proposed model classifies phishing emails based on the entire email content rather than isolated features. Intended to handle with the lack of recent and curated data, a database was labeled using a clustering method that combines KNN and K-Means [17].

Similarly, several research efforts have also focused on applying machine learning techniques to detect Short Message Service (SMS) based phishing attacks [18], [19], [20], [21], we categorized as SMS Phishing Detection (Smishing) in Table 2. Attackers may favor this channel due to its ease of use and the inherent limitations of mobile devices, such as smaller display sizes, that can hinder the detection of subtle phishing indicators. This combination of factors increases the likelihood of successful attacks, as users usually are more prone to quickly interact with SMS messages [18].

A dataset of 32,259 real-world Swahili Smishing messages was collected from mobile money users in Tanzania. Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) models was experimented in this dataset. Synthetic Minority over-sampling Technique (SMOTE) was also explored to address class imbalance in the target class [20]. In [21], a real-world SMS dataset, containing 1,204,057 instances labeled as phishing and not-phishing collected and maintained by the Korean Internet and Security Agency (KISA) between 2018 and 2022 was explored to develop a text evasion attack tool, capable of generating pseudo-variant messages from a given message using an adversarial attack approach [21]. In [18] Mendeley's SMS Phishing and Kaggle's SMS Smishing Collection, both public datasets, is used to developed a Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) model.

Using a different approach, in [19], a machine learning classification of SMS phishing that uses a combined URL characteristics and SMS content analysis was proposed. Rare terms are explored as features utilizing Term Frequency (TF) and the inverse document frequency (IDF) to calculate the TF-IDF score.

Voice phishing (vishing) detection with machine learning, as categorized in Table 2, has also been proposed by leveraging Named Entity Recognition (NER) with Key Tags and sentence-level N-grams in phone call texts in the context of telephone financial fraud. A set of financial counselling texts was collected as a non-phishing dataset, focusing on contexts relevant to voice phishing and tagging with 'Key Tags' to minimize the impact of specific names on classification accuracy, and sentence-level N-grams were used to analyse each sentence within voice phishing texts independently. Using this approach, the study achieved the best performance with SVM and Logistic Regression on the 4-gram dataset [22].

As an alternative approach, a variety of researches explores approaches to URL Phishing Detection, as categorized in Table 2. These works rely on the URL characteristics to

identify potential phishing attacks [10], [11], [12], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37]. A Uniform Resource Locator (URL) is a specific type of Uniform Resource Identifier (URI) used to identify resources on the internet and consist of an important element for web navigation and typically dwell of a scheme (e.g., HTTP, HTTPS) followed by a colon (:) and a sequence of characters. The scheme determines the protocol used to access the resource. HTTPS, a secure version of HTTP, encrypts communication between the client and server, protecting sensitive information such as passwords and credit card numbers. Both HTTP and HTTPS rely on TCP/IP port numbers, with 80 for HTTP and 443 for HTTPS [38].

To further delve into this area, the subsequent subsection presents a comprehensive review of recent research advancements in phishing detection using machine learning techniques. The focus will be on methodologies that leverage URL-based features for enhanced phishing identification.

B. URL-BASED PHISHING DETECTION

1) DATASET AND FEATURES USED IN URL-BASED PHISHING DETECTION

The effectiveness of a classifier is fundamentally determined by the quality and representativeness of the training data. Therefore, rigorous attention to data collection methodologies is imperative, as the performance of machine learning models is inherently linked to the underlying characteristics of the training dataset. This subsection delves into the prevalent data sources and collection techniques employed in recent research on phishing URLs detection with machine learning. How researchers have leveraged public datasets or compiled their own data to accurately reflect the evolving landscape of phishing attacks is the target of our exam.

Some researchers developed their own dataset collection by leveraging public user report services, such as VirusTotal, PhishTank and OpenPhish, to compose the phishing patterns, while benign patterns were selected from popular websites indexed by platforms similar to Alexa Web Information Rankings, Google page rank and Common Crawl [12], [23], [25].

In [23], for instance, a machine learning-based method was developed to classify malicious URLs as whether phishing or malware. The proposed classifier, trained on data self collected VirusTotal URLs incorporated three key feature groups: lexical, hosting, and domain (WHOIS information). Their findings highlight that lexical features shown to be very important to the task, as phishing URLs are more likely to exhibit lexical features that mimic popular brands compared to malware URLs.

In [25], lexical URLs and html content features from website and e-mail are combined to introduce a comprehensive multi-phase phishing detection framework that integrates blacklist filtering, heuristic rule analysis, and machine learning classification using data collected from Siri,

TABLE 2. Categorization of phishing models and detection approaches of recent researches.

Category Key References	Model/Approach/Focus	Description
Phishing Susceptibility Models [7]	Three-stage Phishing Susceptibility Model (PSM)	Proposed by [7] to understand human vulnerability. Categorizes factors into long-term stable (knowledge, personality), situational (stress, workload), and in-the-moment (cognitive effort, email presentation). Aims to guide targeted interventions.
	[15]	Multidimensional Phishing Susceptibility Prediction Model
Website/URL Phishing Detection	Early Detection of Phishing URLs (Current Work)	Focuses on proactively identifying and flagging URLs that mimic specific financial institutions immediately upon registration, with a focus on newly registered and parked domains. This approach integrates brand protection with end-user anti-phishing strategies. It leverages a machine learning-based framework utilizing lexical features for early detection and classification (Current Work).
	[10], [11], [11], [12], [23]–[27], [27]–[37]	URL Characteristic-based Identification
	[25]	Visual or HTML content detection
Email Phishing Detection	Content-based Classification using NLP	Explores application of machine learning techniques (e.g., NLP, LSTM neural networks) to classify email messages as phishing or legitimate based on their entire content. Aim is spam detection or phishing attack prediction.
	[16], [17]	
SMS Phishing Detection (Smishing)	ML-based Classification (CNN, LSTM)	Applies machine learning techniques (Convolutional Neural Network, Long Short-Term Memory) to detect malicious links via SMS by analyzing message content.
	[18], [20]	Combined URL Characteristics and SMS Content Analysis
	[19]	Text Evasion Attack Tool Development
Voice Phishing Detection (Vishing)	Named Entity Recognition (NER) with Key Tags	Proposes a detection process for telephone financial fraud by applying NER with "Key Tags" and sentence-level N-gram analysis on phone call texts. Achieved best performance with SVM and Logistic Regression on 4-gram datasets.
	[22]	

Alexa and PhishTank. The proposed methods employ DNS blacklist checks, content-driven heuristic analysis (based on domain age, URL structure, web traffic metrics, and reputation scores), and a web crawler to dynamically extract relevant features from URLs, websites, and emails. A key contribution of the study lies in its unified approach to evaluating phishing indicators across three vectors—email, website content, and URL structure.

As summarized in Table 3, a variety of studies, for instance [10], [11], [26], [28], [32], [36], [37] rely on public dataset such as UCI Machine Learning Repository, ISCX-URL2016, Kaggle, Mendeley Data and explores methods to improve efficiency on their classifiers. Others, relate to develop their own dataset by combining data to a public dataset with a recent data collection of known phishing URL from an Openfeed (Phishtank) in order to address the need of up-to-date data for an efficient model [10].

Our current work constructed a dataset from four sources to train and evaluate the proposed methods for detecting a domain-specific phishing threats, as shown in Table 3 Real-time SSL certificate monitoring from <https://certstream.calidog.io>, a list containing 2024 most popular domains from <https://radar.cloudflare.com/domains>, a newly registered domains sample collection from <https://whoisds.com>, for positive pattern, and a private dataset of phishing incidents reported by the target company itself.

This decision to utilize a private dataset stemmed from two key factors. Firstly, phishing threats patterns evolve rapidly with changing campaigns, necessitating up-to-date data. Secondly, the project aimed to develop a methodology for detecting phishing threats to a specific organization, rather than a generic phishing pattern detector, requiring a dataset tailored to this objective. The data collection strategy is detailed explained on subsection III-A.

2) DIMENSIONALITY REDUCTION OF URL-BASED PHISHING DETECTION

This subsection provides an overview of various dimensionality reduction methodologies, specifically feature selection and feature extraction, as observed within the scope of the previously selected literature.

High-dimensional datasets pose significant challenges for machine learning models, leading to increased computational complexity and potential degradation of performance. This phenomenon, often referred to as the ‘curse of dimensionality’, needs dimensionality reduction techniques, such as feature selection or extraction, to improve model accuracy and efficiency. These techniques aim to identify and eliminate redundant, irrelevant, and noisy features, while preserving the most informative characteristics of the data [33].

Feature selection techniques can be categorized into five groups: Filter, Wrapper, Embedded, Hybrid, and

TABLE 3. Comparison of approaches on dataset selection and features in recent studies.

research	Data Source	Description	Features
[23]	Virus Total	1.577 Billion URLs collected from 2019-2022 form Virus Total user reports classified into malware or phishing URLs.	Combines Virus Total scanner features with lexical URLs feature and host and Domain information (Whois)
[23], [24]	phishtank.com	The dataset collection consisting of 51,316 legitimate websites and 36,173 phishing websites.	73 lexical URL-based features
[25]	Alexa web information, SiriReputation and PhishTank	Self collection of 200 samples	12 lexical features from URLs combined to website traffic information and domain age
[12]	Phishtank.com and commomcrawl.org	5.1 million URLs collected from Phishtank.com's user phishing report's for the positive phishing label and commomcrawl.org's URLs for benign pattern	Self extracted URLs's lexical features
[10], [26]–[28], [29], [30]	Mohammad & McCluskey's UCI ML Repository	A public dataset containing 11,055 from URLs collected from PhishTank archive, MillerSmiles archive, Google in 2015 labelled into phishing, not-phishing.	30 features set combining lexical URLs patterns with some external information as domain age (Whois), and web traffic or popularity (Google index and Alexa Web Information Services)
[26], [31]	Canadian Institute for Cybersecurity's ISCX-URL2016	114,250 URLs collected from multiple sources data, similar to Openphish, Alexa Web Information was collected and labeled into benign, phishing, spam, malware and defacement was published in 2016.	Combines URLs textual patterns with some external informations as domain age and some html features.
[11], [29], [30]	Mendeley's Vrbancič Phishing-Dataset	Published in 2020, containing 88,647 instances classified into phishing or not phishing, collected from Phishtank.com for positive phishing pattern and from Alexa Web Information's rank for benign pattern.	111 features set, 20 lexical features based on URL, 21 lexical feature based on the domain property, 21 based on filename and also in directory name, 20 features of URL parameters and 16 based on based on resolving URL and external services.
[29], [32], [33]	Mendeley's Tan Phishing Dataset	10,000 phishing webpages a half collected from PhishTank and OpenPhish, for positive pattern, and a half Alexa Web Information's rank and CommonCrawl.org for the benign pattern from 2015-2017.	48 features from URL and web content/source code
[31]–[33], [34]	Mendeley's Hannousse and Yahiouche Dataset	A balanced dataset of 10,430 URLs classified into phishing, not-phishing published in 2021	87 features URL, web content/source code, domain, and page rank
Current Work	Cloudflare radar, cerstream.io, whois.ds combined to a private Three year (2021-2024) real-world user reports from a retail bank	Most accessed sites in 2024 by cloudflare radar index and cerstream.io new certified URLs curated collection for the benign pattern	Utilizes 21 self extracted URLs's lexical features to detect phishing in newly registered domains

Evolutionary. Filter methods, such as Information Gain (IG), Chi-square test, Fisher's score, Correlation Coefficient, Variance Threshold, mean absolute difference (MAD), relief (reliefF, RreliefF), and Dispersion Ratio, rely on intrinsic properties of the data to rank features. Wrapper methods, including Forward Feature Selection (FFS), Backward Feature Elimination (BFE), Exhaustive Feature Selection (EFS), and Recursive Feature Elimination (RFE), evaluate feature subsets based on their performance with a specific model. Embedded methods, such as LASSO Regularization (L1) and Feature Importance, integrate feature selection within the model training process. Hybrid approaches combine multiple feature selection techniques, while Evolutionary-based feature selection utilizes evolutionary algorithms to search for the optimal feature subset [29].

The decision for a feature selection technique also concerns about explainability of models. SHapley Additive exPlanations (SHAP) and embedded feature importance are a widely recommended methods in this context [27], [31], [34]. Feature importance was utilized in a recent a phishing

URL detector using Neural Networks algorithms into a public dataset exploring 30 URL features, geo-IP tracking, and tracker auditing to classify websites as phishing or non-phishing, intended to providing users information about destination and blocking suspicious sites. Their system was validated with real-world phishing simulations using the Go-phish toolkit, demonstrating the real-time detection proposed method [27]. In [30] an approach to feature selection based on feature importance achieves 87.6% reduction in feature quantity with suffering from only a 0.1% deterioration in detecting accuracy, relying on 14 features with best accuracy on phishing detection with Classification And Regression Tree (CART), Random Forest, and Boosting models. SHAP values has been also experimented for feature selection for phishing detection models. SHapley Additive exPlanations (SHAP) are a prominent hybrid approach, reflecting the paradigm shift towards Explainable Artificial Intelligence, centralized in developing more interpretable machine learning (ML) models. Findings in experiments with SHAP demonstrate a performance improvement in recall and

accuracy using with XGBoost and Random Forest model for phishing detection both in lexical URL-based features and also in a feature-set combining URL-based features and HTML content features [31], [34].

Kocyigit et al. employed a Genetic Algorithms (GAs), a class of evolutionary optimization algorithms, for feature selection in [24] in a phishing URLs predicting model achieving suitable results with Random Forest. In the case, the initial 73 feature-set was reduced to 45 features, leading to the best performance and substantial reduction in the computational cost.

The filtering category of feature selection was also explored to improve phishing websites detection with machine learning. Uni-variate feature selection (UFS) method has been also explored in order to mitigate the computational overhead associated with ensemble learning methods on phishing website detection developed with Random Forest, Cat-boost and Gradient Boost, all ensemble decision trees based models, the last ones combining gradient boosting on decision trees. ANOVA F-test was employed within the UFS framework to identify and select the most relevant features, reducing the dimensionality of the data and to improve the efficiency in Cat-Boost and Random Forest leading to a fast and accurate classification [33]. A exploration of a Combined Correlation Features Selection methods achieved suitable accuracy on using a small set of 10 features for phishing detection on two public datasets. This approach uses the feature selection combined the Power Predictive Score (PPS) for identifying correlations between features and the target variable with Recursive Feature Elimination (RFE) for ranking and gradually removing less impacting features. This process ultimately resulted in 10 key features, achieving high classification accuracy for both experimented datasets using Random Forest [32].

Alternatively to Feature Selection, a feature extraction approach, Principal Component Analysis (PCA), was employed by [11], to optimize an XGBoost model, reducing classification time from 1500 ms to 869 ms. PCA is an unsupervised dimensionality reduction technique applicable to supervised learning models, operating by identifying principal components—linear combinations of original features that capture maximum data variance.

Our analysis of recent literature reveals that studies frequently achieve superior performance with a reduced set of features, underscoring the importance of developing automated feature reduction approaches for URL-based phishing detection. We also observed the consistent success of Random Forest and Boosting models, which often attain the highest levels of accuracy, recall, and prediction speed in this domain.

The current work, based on insights from the literature, prioritizes the combination of efficiency and explainability in designing a lightweight and interpretable model for detecting newly registered phishing URLs. Accordingly, we adopt SHapley Additive exPlanations (SHAP) and SelectFromModel, an embed feature importance, as primary

feature selection techniques, given their strong alignment with interpretable machine learning and their successful application in tree-based classifiers such as Random Forest, XGBoost, Gradient Boosting, and Light Gradient Boosting Models. To complement these methods, we also include the Chi-square test, which offers a statistically grounded and transparent approach to evaluating feature relevance. For baseline comparison, we implemented Principal Component Analysis (PCA), a well-known unsupervised feature extraction method.

III. PROPOSED MACHINE LEARNING-BASED FRAMEWORK FOR EARLY DETECTION OF PHISHING URLS IN PARKED DOMAINS

This section presents the proposed framework, which includes data collection, pre-processing, feature selection, model construction and training, and performance evaluation. Figure 1 illustrates the overall process and workflow. The approach begins by monitoring newly registered domains and analyzing them with the machine learning classifier named **Phishing Hunter**,¹ designed to detect phishing attempts. When a domain is classified as phishing and identified as targeting specific organizations, the associated URL is reported to threat intelligence and URL reputation feeds, facilitating integration with antivirus and anti-spam tools. If the phishing attempt involves brand abuse, the organization can escalate the case to the hosting provider, which then reviews the report and initiates actions to deactivate the malicious URL.

Newly registered URLs can be sourced and monitored from platforms such as WhoisDS (<https://whoisds.com>), Whois Data Center <https://whoisdatacenter.org>, and newly issued SSL certificates from CertStream (<https://certstream.calidog.io>). By integrating cybersecurity practices with brand protection strategies, this strategy facilitates proactive measures, including domain takedown and domain reputation bans, to mitigate phishing attempts before they can impact end-users. Figure 2, illustrates the sequential stages to the development of our machine learning model (Phishing Hunter): data collection, pre-processing, feature reduction, data sampling, training, hyperparameter tuning, and evaluation, ensuring robust detection and proactive prevention capabilities.

A. DATA COLLECTION

This section describes the initial phase of the study, illustrated in Figure 2 letter A. Including data collection, cleaning and labeling. The first source included URLs from the CloudFlare Radar ranking of the most-accessed websites globally in 2024 (<https://radar.cloudflare.com>). The second source consisted of URLs obtained from a curated stream of newly issued SSL certificates captured by <https://certstream.calidog.io>. These websites represent a baseline of legitimate traffic. A third source was collected from <https://whoisds.com> free samples

¹<https://github.com/Jaquelinedops/PhishingHunter>

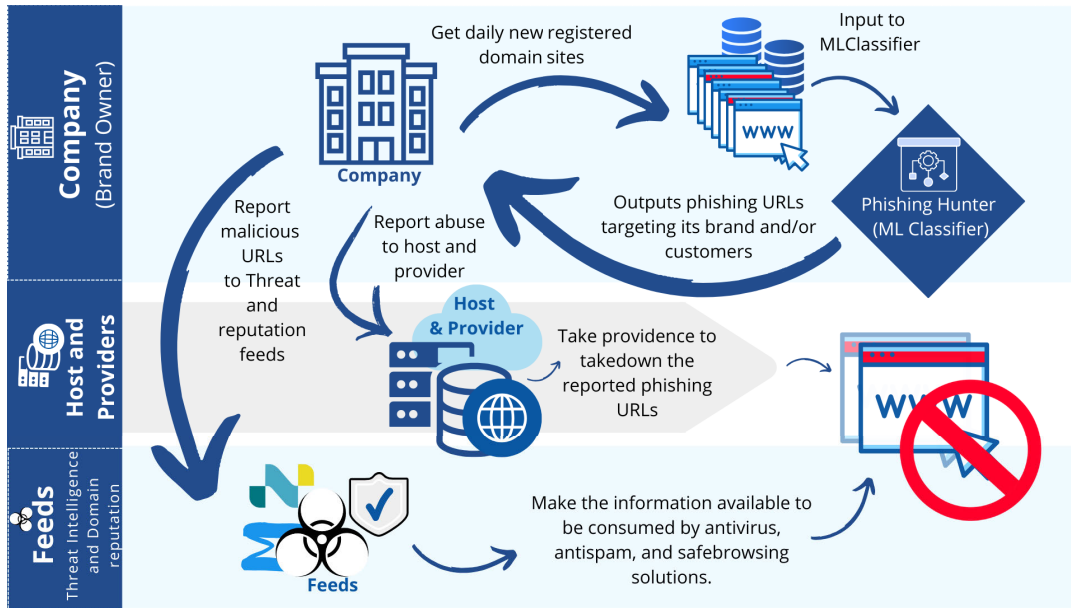


FIGURE 1. Application scenario of the proposed Machine Learning-based framework for Early Detection of Phishing URLs in Parked Domains.

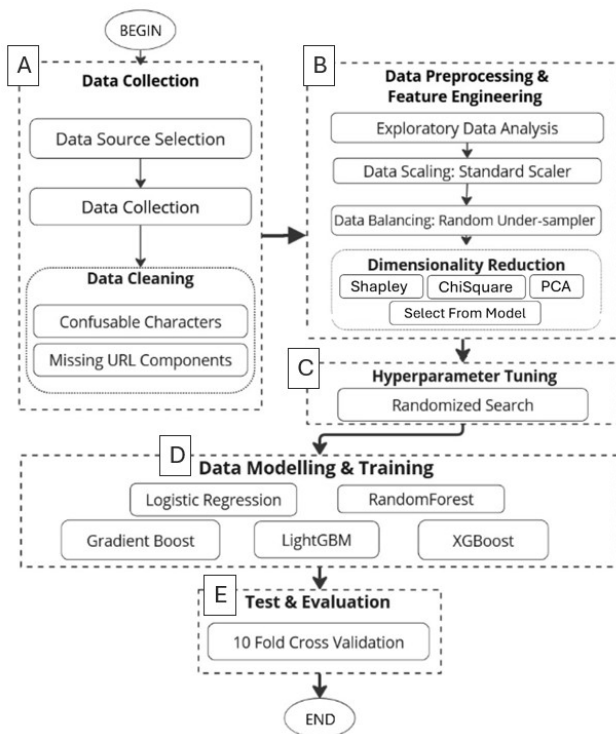


FIGURE 2. Block diagram of the proposed data processing and detection of the proposed Machine Learning based framework for early detection of phishing URLs in parked domains (Phishing Hunter).

of Daily Newly Registered Domains. Finally, the positive patterns, comprising real phishing URLs, were obtained from a private dataset of phishing incidents collected over three years (2021-2024) through curated user reports. All collected

URLs were meticulously labeled as phishing or legitimate and underwent a rigorous curation process to guarantee data fidelity.

B. PRE-PROCESSING AND FEATURE ENGINEERING

This section outlines the data preparation procedures employed in this research. These procedures encompass data transformation, partitioning the data into training and evaluation sets, addressing class imbalance and feature engineering. The primary aim of this phase is to prepare the data for effective model training and evaluation.

During pre-processing, the dataset was divided using stratified random sampling into a training set (80%) and a test set (20%). Due to class imbalance, where one class significantly outsizes another class or classes, the model tends to become biased toward the majority class. To address this issue, techniques such as under-sampling and over-sampling are commonly employed. Under-sampling reduces the majority class instances, matching the minority class, whereas over-sampling generates synthetic samples to increase minority class instances [28]. In this study, we investigated two distinct resampling strategies to address class imbalance: an under-sampling approach and an over-sampling approach based on the Synthetic Minority over-sampling Technique (SMOTE). While over-sampling methods, including SMOTE, generally lead to increased computational training time, they offer the advantage of generate synthetic minority class samples without discarding any majority class (“benign”) instances [39]. This characteristic allowed us to directly assess the impact of reducing majority class information inherent in under-sampling, thereby providing a

more comprehensive understanding of its influence on overall model training and performance.

The first step of feature engineering was conducted by exploring some characteristics related to textual patterns and the basic components of a URL. These components typically include the protocol, sub-domain, domain name, top-level domain (TLD), and subdirectories, as shown in Figure 3.



FIGURE 3. Basic components of URL.

1) DIMENSIONALITY REDUCTION

This section outlines the dimensionality reduction techniques employed in this research. Feature selection was utilized to identify and retain only the most salient features from the original dataset. This process aims to simplify models, mitigate the risk of overfitting, and enhance generalization capabilities. Consequently, feature selection typically yields improvements in predictive accuracy, reductions in training time, and more interpretable model outcomes [40].

Informed by insights from the extant literature, as discussed in Section II-B2, and with a focus on balancing computational efficiency and model explainability for a lightweight phishing URL detection model, we selected SHapley Additive exPlanations (SHAP) and SelectFrom-Model (embedded feature importance). These methods align strongly with principles of interpretable machine learning and have demonstrated efficacy with tree-based classifiers. Furthermore, the Chi-square test was incorporated due to its statistically robust and transparent approach to evaluating feature relevance. For baseline comparison, Principal Component Analysis (PCA), an unsupervised dimensionality reduction method, was implemented.

Select from Model is a feature selection based on model importance scores. This technique involves training a suitable model on the dataset. The trained model then provides importance scores for each feature, revealing its contribution to the model's predictions. A target threshold is defined, and features with importance scores exceeding this threshold are selected for subsequent analysis or model training.

We applied a threshold derived from a sensitivity analysis of feature importance scores, as computed by the underlying estimator (e.g., Random Forest or Logistic Regression with L1 regularization). Thresholds based on the mean and median importance values were explored to establish a data-driven criterion for feature selection.

An alternative feature selection approach was also experimented based on **SHapley Additive exPlanations (SHAP)**, specifically, the SHAP tree explainer, a variant of SHAP designed for tree-based models, was adopted. It is based on the concept of Shapley values from game theory and assigns a “score” to each feature, showing its contribution to the final prediction and is traditionally in model interpretability technique for machine learning that focus on how features influence a model's predictions [41]. SHAP identifies if each feature's contribution is positive or negative and provide global and local interpretability [42]. SHAP was also assessed with a sensitivity analysis varying in threshold within a pre-defined range. We evaluated cutoffs at range [0.005, 0.01, 0.05, 0.1].

For the third approach, the **Chi-Squared** (χ^2) statistical test was employed to evaluate the dependency between each feature and the target variable. This test measures whether the observed frequencies of feature-class combinations significantly differ from frequencies expected if the feature and target variable were independent [43]. The Chi-Squared statistic (χ^2) is mathematically defined as:

$$\chi^2 = \sum_{k=1}^n \frac{(O_k - E_k)^2}{E_k} \quad (1)$$

where:

- O_k is the observed frequency of occurrences for category k ,
- E_k is the expected frequency for category k assuming independence between the feature and the target variable,
- n is the total number of classes.

Higher χ^2 values indicate greater dependency between the feature and the target variable. Conversely, lower values indicate independence. The test produces a p -value that reflects the probability of observing the calculated χ^2 under the assumption of independence. Features with low p -values (typically $p < 0.05$) demonstrate statistically significant associations with the target and are considered relevant for feature selection. For this approach, we retained features that achieved p -values < 0.05 , following conventional statistical significance criteria. This ensures that retained features have a statistically meaningful association with the target variable, reducing the risk of including irrelevant predictors.

A **Principal Component Analysis (PCA)** approach was also experimented. PCA is an unsupervised learning algorithm widely utilized for dimensionality reduction in supervised models. This technique transforms the original data into a new set of uncorrelated variables, known as principal components [1]. The central component of PCA is the covariance matrix C , defined as:

$$C = \frac{1}{n-1} XX^T \quad (2)$$

where the diagonal terms of matrix C represent the variance of individual features, while the off-diagonal terms quantify

the covariance between pairs of features. PCA transforms the original features to eliminate these covariance terms, resulting in a diagonal covariance matrix C_{PCA} . The diagonalization of matrix C is achieved using eigenvectors:

$$C_{PCA} = \frac{1}{n-1}YY^T \Rightarrow \frac{1}{n-1}(PX)(PX)^T \quad (3)$$

In this transformation, the rows of P correspond to eigenvectors derived from the covariance matrix XX^T , and $Y = PX$ represents the transformed data in the new feature space defined by the principal components.

The PCA covariance matrix C_{PCA} calculates variance along principal components by rearranging diagonal entries according to the directions of highest variance in the dataset. Dimensionality reduction occurs by discarding principal components associated with lower variance. Thus, the dimension of the transformation matrix P is $(k \times p)$, where k is the number of retained principal components and p is the original number of features [11].

C. HIPERPARAMETER TUNNING

A pipeline including the Randomized Search was configured to evaluate 10 randomly selected combinations of hyperparameters ($n_{\text{iter}} = 10$) from a predefined search space. Each configuration was assessed using 10-fold cross-validation ($cv = 10$), to search for the optimal hyperparameter set in order to prioritize best recall and accuracy model scores. This approach conducts random sampling from predefined hyperparameter distributions, evaluating a subset of possible parameter combinations. Compared to exhaustive search, it reduces computational demands while efficiently covering the hyperparameter search space [44]. For XGBoost, Random Forest, and LightGBM models, hyperparameter grid were selected based on theoretical influence on model complexity, regularization, and generalization [45], [46], [47]. To mitigate overfitting, particularly, a small maximum depth of trees was guaranteed to all models. Additionally, L1 (reg_alpha) and L2 (reg_lambda) regularization were configured for the boosting models (XGBoost, LightGBM), while for Random Forest, max_features was set to a small value to improve variance and reduce correlation among trees. The parameter grid is related in Table 4.

D. MODELING AND TRAINING

In this section, we present the third stage of the framework, where the selected machine learning models are created and trained using the prepared and reserved subset of data from the previous stage.

Logistic regression served as our baseline model, providing a benchmark for comparing the performance of more complex models. The baseline helps determine whether models with greater complexity yield significant improvements over a simpler approach. Subsequently, ensemble models including Random Forest, XGBoost, LGBM, and Gradient Boosting were initially executed using their default parameters. These ensemble models were selected based on

TABLE 4. Hyperparameter configuration for model tuning.

Model	Hyperparameters and Values
XGBoost	$n_{\text{estimators}}$: 50, 100, 200, 300, 500 learning_rate: [0.01, 0.6] subsample: [0.3, 0.9] max_depth: 4, 6, 8, 10 reg_alpha, reg_lambda: 0, 0.1, 0.5, 1
Random Forest	$n_{\text{estimators}}$: 50, 100, 200, 300, 500 criterion: gini, entropy max_depth: 4, 6, 8, 10 min_samples_split: 2, 5, 10 min_samples_leaf: 1, 2, 4 max_features: sqrt, log2
LightGBM	$n_{\text{estimators}}$: 50, 100, 200, 300, 500 learning_rate: [0.001, 0.01, 0.05, 0.1] num_leaves: 30, 50, 70, 90 max_depth: 4, 6, 8, 10 min_child_samples: 20, 50, 100 reg_alpha, reg_lambda: 0, 0.1, 0.5, 1

their effectiveness demonstrated in related works reviewed in the literature.

E. EVALUATION

This section outlines the evaluation metrics employed to select the most suitable classification model for the proposed task. Model performance was assessed using the metrics **accuracy**, **recall**, **precision** and **F1-score**. These metrics provide complementary insights, especially in scenarios involving imbalanced datasets.

Accuracy measures the overall correctness of a classification model by calculating the ratio of correctly predicted instances (true positives and true negatives) to the total number of instances in the dataset. The accuracy metric is mathematically expressed by Equation 4:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

where TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives. Accuracy is suitable for balanced datasets but can be misleading if the dataset is significantly imbalanced, as the majority class can dominate the metric.

Recall, also known as sensitivity or true positive rate (TPR), measures the proportion of actual positive instances correctly identified by the model. Recall quantifies the model's capability to avoid false negatives and is specifically important in contexts where missing actual positives carries expressive risk, such as fraud detection or security-related tasks. The mathematical expression for recall is provided by Equation 5:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

Precision measures the proportion of positive predictions made by the model that are correct. This metric is particularly

important when the cost of false positives is high, reflecting the reliability of the positive predictions. Precision is calculated as shown in Equation 6:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

The **F1-score** provides a balanced measure between precision and recall, combining both metrics into a single value. It is useful for evaluating model performance, especially in datasets with imbalanced class distributions, as it considers both false positives and false negatives simultaneously. The F1-score is defined by Equation 7:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

Accuracy and recall were selected as priority metrics to assess the performance of the classification models also for hyperparameter stage. The emphasis is justified by the nature of the problem, where detecting all malicious URLs is primary to protect users. A false negative in this context could lead to more severe consequences, such as data loss or malware infections.

To ensure a reliable evaluation of the selected metrics, 10-fold cross-validation was applied. In cross-validation, a dataset is divided into multiple subsets; the model is repeatedly trained on some subsets, and its performance is validated on the remaining subset. This method provides robust estimates of model effectiveness and reduces the risk of overfitting or biased performance estimates.

IV. RESULTS

This section presents the results of XGBoost, LGBM, Gradient Boosting, and Linear Regression classifiers in detecting phishing URLs and is organized into subsections: Data Collection, Data pre-processing and Feature Engineering, Modeling and Training, Hyperparameter Tuning, Dimensionality Reduction, and Test and Evaluation. Each subsection discusses the results obtained from the respective stages of the framework described in Figure 2.

A. DATA COLLECTION

A total of 211,659 URLs were collected from diverse public sources: Callidog Certstream [48] for real-time SSL certificate monitoring, Whois Domain Search [49] for daily registered domain sample, Cloudflare Radar [50] for a list of most accessed domains, and a private database from the target company's user-reported phishing incidents. The distribution of these URLs across classes is detailed in Table 5.

TABLE 5. Data source distribution by category.

Source	URLs	Category
Top Accessed Domains by Cloudflare	50,000	False (Benign)
Certstream	70,034	False (Benign)
Newly Registered Domains	50,000	False (Benign)
Phishing	41,625	True (Phishing)

B. DATA PRE-PROCESSING AND FEATURE ENGINEERING

Data pre-processing tasks was conducted to ensure data quality and consistency. This stage encompassed handling missing values, labeling data, and standardizing character encoding. Additionally, a novel approach was implemented to address confusable Unicode characters. These characters, visually similar and easily confused with others, were replaced with their corresponding Latin equivalents. This step aimed to enhance model robustness by mitigating the risk of evasion techniques employed by malicious actors who exploit confusable characters to bypass detection mechanisms.

The feature engineering step was proceed based on findings from the literature reviewed in the Section II-B. Primarily, focused on extracting lexical characteristics from each element of URL's basic components. These features included: TLD length, number of directories, length of the first directory, hostname length (calculated as the sum of sub-domain, domain name, and TLD lengths), and the protocols (https) presence, sub-domains, domain names, and TLDs. Furthermore, the presence of "Sponsored TLDs", as defined by ICANN's Consensus Agreements for New Generic Top-Level Domains (gTLDs) as those serving specific community interests, was identified as a relevant feature. Examples of Sponsored TLDs include ".gov", ".edu", ".aero", and ".mil".

Other features were also extracted from the URL's textual pattern, including the frequency of special characters, the total number of letters, and the number of digits. Additionally, the frequent_terms feature was built from the positive samples (phishing URLs that target the financial institution). This feature represents the terms/tokens that cumulatively account for 40% of the total frequency in the dataset.

It was obtained by first tokenizing the domain and path components of the phishing URLs using non-alphanumeric delimiters. These tokens then underwent pre-processing, which involved the removal of stop words. Following this, a stemmer (RSLPStemmer) was applied to reduce terms to their root forms, thereby normalizing the vocabulary. Finally, a cumulative frequency analysis was performed on these processed terms, identifying and retaining the subset of terms that collectively contributed to the top 40% of the total term frequency, resulting in 50 terms/tokens. Finally, a total of 20 features were extracted, as detailed in Table 6.

After building an initial feature set, the dataset was split into training (80%) and testing datasets (20%). To ensure a representative distribution of classes in both the training and testing sets, we employed stratified sampling, preserving the original class proportions in each subset. Data was randomly shuffled before splitting and a random seed set was defined to ensure reproducibility of the experiments.

C. MODELING AND TRAINING

Logistic Regression, XGBoost, LGBM, Gradient Boosting, and Random Forest models were first trained on this train data subset. At this stage, the model evaluation was conducted

TABLE 6. Feature set description.

Feature	Type	Description
abnormal_url	Categorical	Indicates if the URL deviates from standard patterns by missing common components.
contains_https	Categorical	Indicates the presence of Hypertext Transfer Protocol Secure (HTTPS) in the URL.
contains_ip	Categorical	Indicates if the URL contains an Internet Protocol (IP) address in its string pattern.
contains_shortner	Categorical	Indicates if the URL contains a known URL shortening service (e.g., bit.ly).
contains_sTLD	Categorical	Indicates if the URL contains a sponsored top-level domain (sTLD)(e.g., .edu, or .mil)
frequent_terms	Continuous	Indicates if the URL features phishing terms commonly targeting the company.
quantity_@	Continuous	Count of '@' characters present in the URL.
quantity_%	Continuous	Count of "%" characters in the URL
quantity_ww	Continuous	Count of occurrences of the substring "ww"
quantity_=_	Continuous	Count of "=" characters in the URL
quantity_-	Continuous	Count of "-" characters in the URL
quantity_slash	Continuous	The count of "/" characters within the URL, representing the number of subdirectories.
quantity_?	Continuous	Number of "?" characters in the URL, indicating the presence of a query string or parameters.
quantity_numbers	Continuous	Count of numeric characters in the URL
length_first_directory	Continuous	Measures the length of the first directory component in the URL.
length_letters	Continuous	The total count of alphabetic characters within the URL.
length_url	Continuous	Total length of the URL string.
length_.	Continuous	The count of "." characters within the URL.
length_hostname	Continuous	The count of characters within the hostname.
length_tld	Continuous	The length of the top-level domain (TLD).

using 10-fold cross-validation, focusing primarily on recall performance. Logistic Regression served as the baseline model, providing a comparative benchmark for assessing the performance gain achieved by the more complex ensemble methods (Random Forest, XGBoost, LGBM, and Gradient Boosting). This comparative analysis aimed to determine whether the increased complexity of these ensemble models translated into a statistically significant improvement in phishing URL detection accuracy.

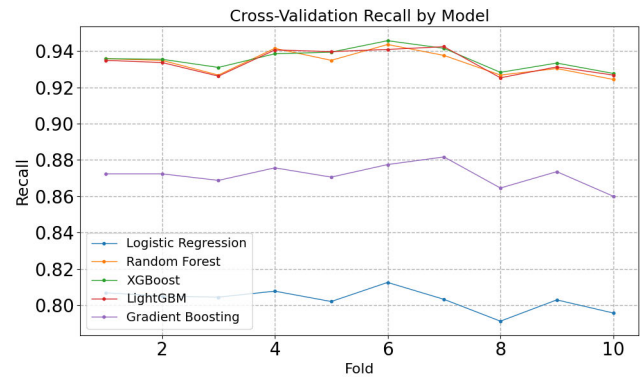


FIGURE 4. Comparison of model recall using 10-Fold cross validation.

As shown in Table 7, XGBoost and LGBM achieved the highest mean recall value with (93.56%) and (93.41%) at this stage. Followed by Random Forest and Gradient Boosting models, achieving 93.36% and 87.17% respectively. Logistic Regression underperformed achieving 80.32% recall.

As this research handles an imbalanced dataset, another training session was evaluated using an approach based on Random under-sampling technique utilizing a majority strategy. This approach reduced the majority class in the training set to 33,300 examples, resulting in a balanced dataset with two equal-sized classes, positive and negative.

To delve deeper into finding the best-performing model, a new round of tests within the five models was realized. As illustrated in Figure 5, in the second test session using the balanced dataset all 5 models exhibited a expressive performance enhancement in recall compared to the results obtained using the initial training dataset (Figure 4).

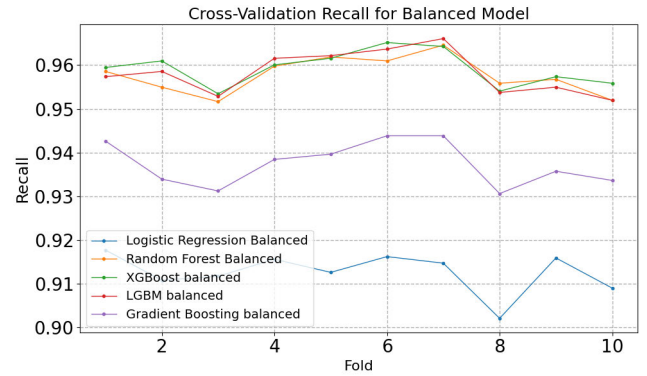


FIGURE 5. Recall comparison across models using 10-Fold cross-validation (Balanced dataset).

By analyzing Figure 5, is possible to observe that the ensemble methods, namely XGBoost, LGBM, Random Forest, and Gradient Boosting, consistently outperform Logistic Regression also in the balanced dataset. XGBoost achieved the best performance with a mean recall of 95.92%, followed closely by LGBM at 95.83% and Random Forest at 95.77%. Gradient Boosting achieved a mean recall 10-fold cross validation of 93.74%. In contrast, Logistic Regression achieves a mean recall of 91.26%, indicating a lower capability in identifying positive instances compared to the ensemble methods in this scenario. To evaluate the impact of sampling strategies on model performance, we compared under-sampling and SMOTE across multiple classifiers. Results indicate that SMOTE had a marginal improvement

in accuracy, particularly for gradient boosting and LGBM models, though the improvement was marginal (e.g., 96.48% to 96.48% for gradient boosting, and 96.47% to 97.60% for LGBM). However, SMOTE also introduced greater training overhead due to synthetic data generation. Notably, under-sampling consistently led to the highest recall scores across all models, reaching up to 95.92% with XGBoost.

Table 7 presents the accuracy and recall scores of all 5 machine learning models first trained on a balanced and then unbalanced dataset. It is possible to note that Ensemble methods, Gradient Boosting, LGBM, XGBoost, and Random Forest, demonstrated higher accuracy compared to Logistic Regression and both test sessions. While balancing the dataset significantly improved recall for all models, it had a minor impact on accuracy, with some instances showing slight decreases. These findings suggest that, for this scenario, ensemble methods offer robust accuracy, and that class balancing is primarily beneficial for improving recall, especially for models more susceptible to class imbalance, such as Logistic Regression.

TABLE 7. Performance comparison of models training (mean 10-fold cross-validation).

Model	Recall (%)	Accuracy (%)
Logistic Regression	80.32	94.99
Logistic Regression Under-sampled	91.26	92.31
Logistic Regression SMOTE	91.26	92.40
Gradient Boosting	87.17	96.48
Gradient Boosting Under-sampled	93.74	95.17
Gradient Boosting SMOTE	93.82	96.48
LGBM	93.41	97.67
LGBM Under-sampled	95.83	96.47
LGBM SMOTE	95.54	97.60
Random Forest	93.36	97.75
Random Forest Under-sampled	95.77	96.47
Random Forest SMOTE	95.28	97.36
XGBoost	93.56	97.79
XGBoost Under-sampled	95.92	96.58
XGBoost SMOTE	94.17	97.38

D. HYPERPARAMETER TUNNING AND DIMENSIONALITY REDUCTION

As shown in Table 7, XGBoost, Random Forest, and LGBM demonstrate stronger performance on the balanced dataset, achieving high accuracy and recall with comparable results across both metrics. These models were selected for further analysis, including hyperparameter tuning and the application of dimensionality reduction techniques. At this stage, the training phase within the dimensionality reduction employed a pipeline for hyperparameter optimization, using the parameters described in Section III-C, to ensure optimal performance on the selected feature subsets while minimizing the risk of overfitting.

The randomized search for all models lead to an improvement in comparison to the prior results shown on Table 7, with LGBM Model within the full future set achieving **97.28%** and a recall of **96.02%**. While, XGBoost and Random Forest, achieved 95.71% and 95.82%, respectively.

Confusion matrix in Figure 6 provides a detailed breakdown of the LGBM model's performance on the phishing classification task. The model achieved 33,186 true positives and 7,994 true negatives, indicating its ability to correctly identify both non-phishing and phishing URLs. The model also produced 331 false negatives and 821 false positives. After optimizing the hyperparameters in the full-set features model, further analysis was conducted to identify the most suitable subset of features by applying dimensionality reduction techniques. Some feature selections approach was employed using a machine learning pipeline consisting of feature selection using a XGBoost, Random Forest classifier and LGBM. The pipelines were trained on the balanced dataset. The result was evaluated using 10-fold cross-validation with recall as the scoring metric.

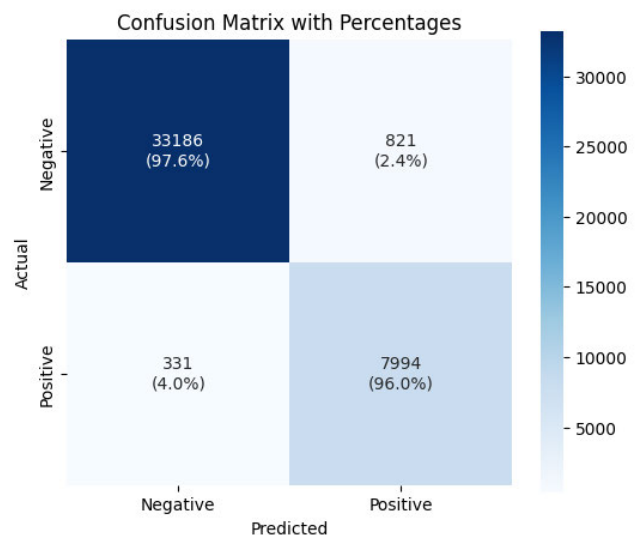


FIGURE 6. Confusion matrix for LGBM 20 features' model performance.

Figure 7 illustrates the feature importance scores obtained using SelectFromModel within a decision tree model's (Random Forest) embed score function. The analysis reveals that 'frequent_terms', 'quantity(.)', 'contains_https', 'url_length', 'quantity_letters', 'length_tld', 'quantidade_fd' are the most influential features in predicting phishing URLs. Notably, features such as 'abnormal_url', and 'quantity of ', 'contains_ip' and 'contains_shortner' exhibited minimal impact on the model's performance, suggesting that their predictive power is limited in this context.

Best results was achieved by utilizing the median importance threshold a feature set with 10 remaining feature was selected for a new train and test session pipelining ('frequent_terms', 'quantity(.)', 'quantity(?)',

'url_length', 'length_hostname', 'length_fd', 'length_tld', 'quantity_numeric', 'quantity_letters', 'contains_https').

The result grouped by Feature Importance is shown in Figure 8. LGBM achieved the best recall with 93.91% and the Time elapsed to predict was reduced in 78% compared to the full set LGBM optimized model (from 141,22 to 28.24 milliseconds), achieving 95.20% accuracy. XGBoost achieved the best recall 96.01% and F1-score (90.17%) and also 93.09% recall, with a reduction of 78.97% on the time elapsed to predict. In this scenario, Random Forest underperformed in all metrics, achieving 91.95% recall, but also had a reduced time to predict from 1595.09 milliseconds to 378.35.

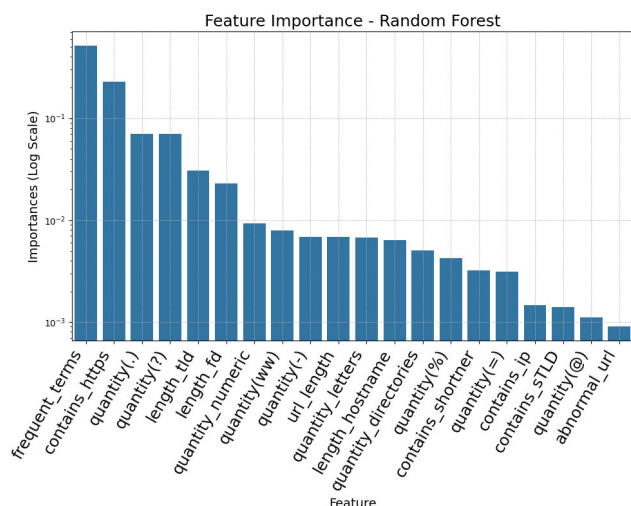


FIGURE 7. SelectFromModel-based feature importance ranking.

A second approach to feature selection was also applied, by utilizing Chi-Square. Figure 8 illustrates the Chi-Square measure of the score for each feature. The analysis reveals that 'url_length', 'quantity_letters', 'quantity_numeric', and 'length_fd', 'frequent_terms' are the most influential features in predicting phishing URLs. Features such as 'abnormal_url', and 'contains_sTLD' and contains_shortener, contains_ip and quantity_(@) exhibited less impact on the model's performance, suggesting that their predictive power is limited in this context. The features 'abnormal_url' and 'contains_sTLD' demonstrate a p_value higher than 0, what means they have no significant dependence on the target label and was discarded, resulting in an 18-feature set.

The three models was retrained and tested with the chi-square result feature set, the results can be visualized in the 8, LGBM achieved the best scores in all metrics, with 96.01% recall, almost achieving the same score from LGBM full set optimized model (96.02%). Accuracy, Precision and F1-score achieved by LGBM model was slightly superior to the Full feature set, showing that the 18-feature set is also suitable. By the way, no improvement was observed in time to predict.

XGboost model experimented slight improvement in recall(0.28%) and accuracy (0.15%) related to the full feature set.

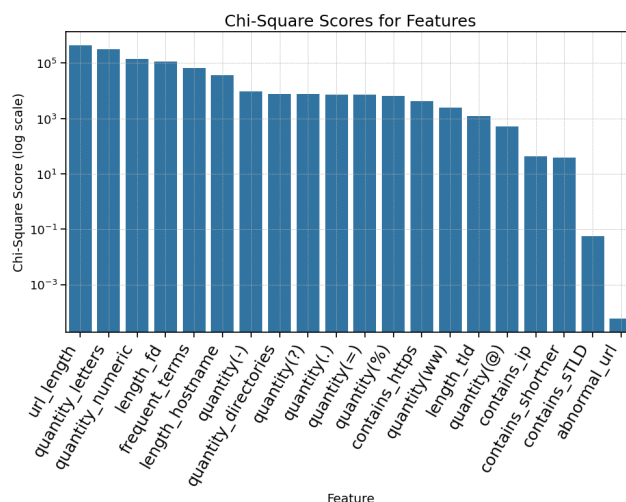


FIGURE 8. Feature evaluation using chi-square test.

Another approach for feature selection was also applied by applying SHAP. The Figure 9 presents the SHAP summary plot. Features in the model are listed on the y-axis, ordered by their importance (top features have the most significant impact on the model's predictions). The x-axis represents the SHAP values, which measure the impact of each feature on the model's output. Positive SHAP values (to the right of zero) indicate that the feature pushes the prediction higher, while negative SHAP values (to the left of zero) lower the prediction. The spread of dots shows the range of SHAP values for each feature. A wide spread indicates that the feature has varying effects on predictions across different observations.

A SHAP decision plot is shown in Figure 10, which illustrates the cumulative contribution of features to the Random Forest best model's predictions for multiple observations. The features of the model are listed along the y-axis, ranked by their importance. Top feature, frequent_terms, has the most influence on predictions, while the bottom features have less impact. The x-axis represents the model's output value (predicted probability). The value increases as the features cumulatively push the prediction higher. Each line represents the path of a single observation as the contributions of features are added cumulatively from top to bottom. The color of the lines also works by corresponding to the feature values for each observation, where blue means low feature value and red, high feature value.

Figure 9 represents the SHAP value impact on model output in the train set, and model output value respectively. By analyzing both figures, it's possible to note that abnormal_url, contains_ip, contains_shortener and contains(@) and quantity(%) has low influence to the model. Following the sensitivity analysis, an optimized 14-feature subset was

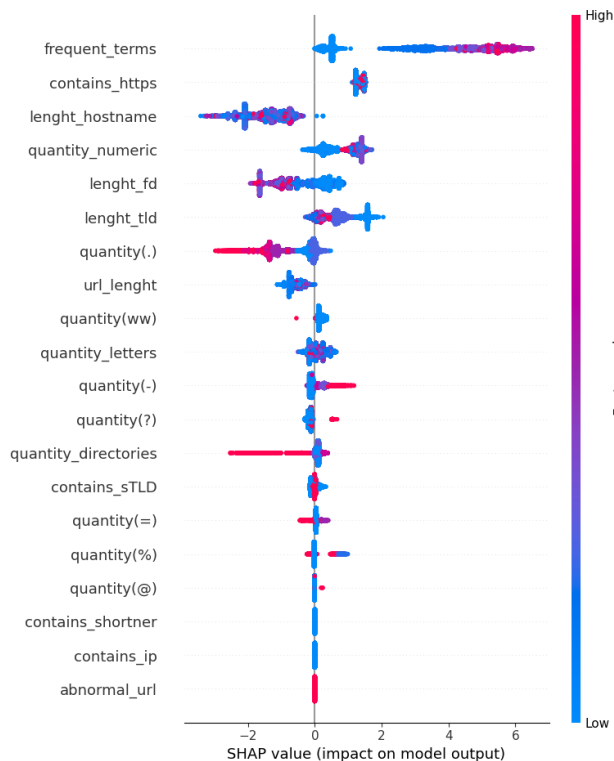


FIGURE 9. Shap summary plot.

identified by applying a 0.05 threshold to the mean absolute SHAP values. Accordingly, to Table 8, a significantly lower recall than obtained using the full feature set (96.02%), although this reduced feature space can be favorable to achieve time performance, by reducing 22% the time to predict compared to the full set optimized LGBM model.

In this scenery, Random Forest had a marginal improvement in recall, surpassing the full-feature model by 0.01% using the full feature set and 50% reducing the time elapsed to predict while keeping the potential benefits to improved model interpretability. XGBoost also reduced prediction time by 81%, although it resulted in a 0.05% decrease in recall.

Principal Component Analysis (PCA) was applied to extract essential information from the data while removing less relevant or redundant features. The approach aimed to reduce data complexity by selecting the minimum number of components needed to explain at least 95% of the total variance. The results obtained from the test set are reported in Table 8.

Models trained on the transformed data (principal components) exhibited a decrease in recall compared to other approaches. However, this method led to improved accuracy across all models and achieved strong F1-scores, reflecting a balanced trade-off between precision and recall.

E. TEST AND EVALUATION

In this subsection, results of the model in the test stage are discussed in relation to the chosen metrics. The results

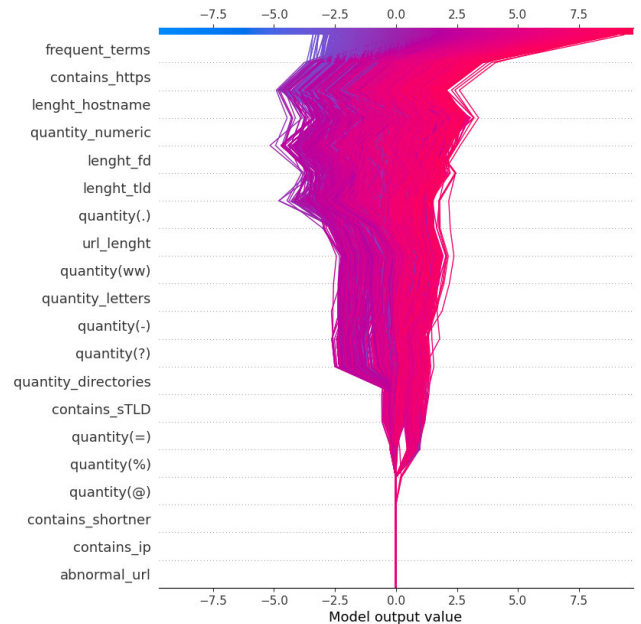


FIGURE 10. Shap decision plot: cumulative contribution of features.

TABLE 8. Test model performance of feature selection.

Model	Recall	Accuracy	Precision	F1-Score	Time(ms)
Feature Importance					
Random Forest	91.95	95.41	85.72	88.73	353.57
XGBoost	93.09	96.01	87.42	90.17	15.85
LGBM	93.91	95.20	83.67	88.49	27.72
PCA					
Random Forest	92.98	97.69	95.21	94.08	443.21
XGBoost	93.30	97.78	95.36	94.32	70.15
LGBM	93.31	97.73	95.06	94.18	31.89
SHAP					
Random Forest	95.72	97.19	90.54	93.06	733.31
XGBoost	95.77	97.28	90.88	93.26	16.42
LGBM	95.95	97.29	90.79	93.30	101.51
Chi-Square					
Random Forest	95.72	97.16	90.38	92.98	757.16
XGBoost	95.99	97.19	90.33	93.08	26.38
LGBM	96.01	97.29	90.75	93.31	112.99
Full feature set					
Random Forest	95.71	97.16	90.40	92.98	1479.46
XGBoost	95.82	97.31	91.00	93.35	87.74
LGBM	96.02	97.28	90.69	93.28	141.22

obtained from the model testing stage are presented in Table 6. The LGBM model trained with the complete feature set achieved the highest recall and accuracy. However, models employing Chi-Square and SHAP feature selection demonstrated comparable performance while substantially

reducing prediction time, while keeping the explainability of the models. Among all evaluated models, XGBoost trained with ten importance-selected features exhibited the lowest computational overhead, performing 42,332 predictions in 15.85 milliseconds. Additionally, LGBM trained with five selected features showed superior accuracy and recall, combined with moderate prediction speed, indicating suitability as the optimal model for the dataset used in this study.

V. CONCLUSION

This research presents a novel framework encompassing data collection, modeling, and application scenarios for the early detection of phishing URLs. This approach leverages machine learning techniques to identify suspicious URLs even when they are in a parked domain stage, thereby assisting anti-fraud and brand protection preventive initiatives.

Our method for collecting and preparing the training dataset achieved high efficiency in extracting relevant features and identifying significant patterns. The proposed approach was successfully applied to the specific environment of the Brazilian bank, showcasing its suitability and robustness.

The LGBM model, utilizing a 20-feature set, yielded the best performance, achieving 96.02% recall and 97.28% precision. Furthermore, the implementation of feature selection using SHAP (14-feature set) and Chi-Square (18-features set) led to a lightweight model with minimal performance loss, making it suitable for practical deployment, while keeping their explainability. With the 14-feature set reducing more than 80% (XGBoost) of time to predict, with 0.05% loss in recall and 0.03% loss in accuracy.

To address the dynamic nature of phishing threats, it is important to recognize that the effectiveness of the detection model may degrade over time as new phishing techniques and patterns emerge. Therefore, it is recommended to retrain the model using up-to-date data periodically to ensure its accuracy and relevance. At the observed institution, model retraining for phishing threat detection was experimentally conducted every four months. This provided an effective balance between adaptability to evolving threats and resource efficiency. Further evaluation is recommended to determine the best retraining period.

While this research presents a promising framework for phishing detection, we acknowledge a current limitation: the absence of established benchmarks explicitly designed for brand protection and early detection. This, particularly as our proposed framework encompasses the collection of private and brand-specific data, precluded direct comparative analysis. Consequently, recommendation for future works involves validating this approach through extensive experimentation with datasets from other companies, including financial institutions and e-commerce. Such testing enable a comprehensive assessment of the framework's generalizability, by analyzing the performance through different scenarios, contributing to a broadly applicable phishing prevention solution.

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